



 Unit 1 – Thinking Geographically	 Unit 2 – Population & Migration	 Unit 3 – Cultural Patterns & Processes
• Maps: reference (political, physical) vs thematic (choropleth, dot, proportional symbol, isoline, cartogram); mental maps show perception • Geographic data: quantitative (statistics, census) vs qualitative (interviews, fieldwork) • Geospatial tech: GPS (location), GIS (layered analysis), remote sensing (satellite/aerial imagery); influences urban planning, disaster response, redistricting • Location: absolute (coordinates, addresses) vs relative (relation to other places) • Place: physical (climate, terrain) + human (culture, economy) characteristics • Spatial concepts: distribution, density, concentration, pattern; distance decay (interaction ↓ as distance ↑); time-space compression (tech reduces effective distance) • Human-environment interaction: environmental determinism (environment shapes culture) vs possibilism (humans adapt environment); sustainability • Scales of analysis: local, national, regional, global; patterns and trends differ at different scales • Regions: formal (uniform trait, e.g., Corn Belt), functional (organized around node, e.g., metro area), perceptual/vernacular (based on perception, e.g., "the South")	• Population distribution: ecumene = inhabited areas; influenced by climate, water, topography, resources • Density: arithmetic (total pop ÷ total area), physiological (pop ÷ arable land), agricultural (farmers ÷ arable land) • Population composition: age-sex pyramids show structure; dependency ratio = (youth + elderly) ÷ working-age • Key rates: CBR, CDR, NIR = CBR – CDR, TFR, IMR; doubling time ≈ 70 ÷ NIR • DTM: Stage 1 (high birth/death), 2 (death falls), 3 (birth falls), 4 (low both), 5 (birth < death) • Epidemiologic Transition: disease patterns shift with DTM — infectious → chronic/degenerative • Malthusian theory: pop grows geometrically, food arithmetically → crisis; critiqued for ignoring technology • Population policies: pronatalist (incentivize births) vs antinatalist (limit births, e.g., China's one-child policy) • Women & demographic change: female education/employment → lower TFR, later marriage, smaller families • Aging populations: rising dependency ratio, pension/healthcare strain; common in Stage 4–5 countries • Push factors: conflict, famine, persecution, disaster; pull factors: jobs, safety, family, education • Migration types: voluntary (economic, chain, step), forced (refugees, IDPs, asylum seekers); guest workers • Effects: brain drain/gain, remittances, cultural diffusion, demographic shifts; Ravenstein's laws describe patterns	• Culture: traits, complexes, systems; folk (traditional, local, slow diffusion) vs popular (mass media, rapid global spread) • Cultural landscape: built environment (architecture, land use, signs, memorials) reflects beliefs, values, identity • Language: families (Indo-European = largest), lingua franca, dialects, isoglosses; language endangerment from globalization • Ethnicity: identity tied to ancestry, territory; ethnic enclaves, ethnic neighborhoods; gender roles vary by culture • Types of diffusion: relocation (migration); expansion — contagious (person-to-person), hierarchical (top-down, e.g., cities first), stimulus (idea modified and adopted) • Historical causes: colonialism, trade routes (Silk Road), conquest, religious missions spread culture globally • Contemporary causes: technology, social media, globalization, transnational corporations, mass transportation • Universalizing religions: actively seek converts — Christianity, Islam, Buddhism; spread by expansion + relocation diffusion • Ethnic religions: tied to place/ethnicity — Hinduism, Judaism, animism; spread mainly by relocation diffusion • Effects of diffusion: acculturation (adopt dominant traits), assimilation (original culture replaced), syncretism (blend of cultures), multiculturalism; cultural convergence vs divergence
 Unit 4 – Political Patterns & Processes	 Unit 5 – Agriculture & Rural Land-Use	 Unit 6 – Cities & Urban Land-Use
• State: defined territory, permanent population, sovereignty, govt; nation: shared cultural identity • Nation-state (Japan), stateless nation (Kurds, Palestinians), multinational state (Russia), multistate nation (Koreans) • Political processes: colonialism, decolonization, neocolonialism shaped modern political boundaries • Territoriality: state control of territory; shatterbelt = contested region between rival powers • Boundary types: antecedent (before settlement), subsequent (after culture), superimposed (imposed externally), relic (no longer functions) • Boundary process: defined → delimited → demarcated → administered; disputes: definitional, locational, operational, allocational • Internal boundaries: voting districts; gerrymandering = manipulating district shapes for political advantage • Governance: federal (power shared — US, India) vs unitary (centralized — France, China); affects spatial organization • Devolution: state power transferred to local/regional level; caused by ethnic separatism, economic inequality, physical geography • Challenges to sovereignty: supranational orgs (EU, UN, NATO, WTO), terrorism, globalization • Centrifugal forces (divide): ethnic conflict, economic inequality; centripetal forces (unite): nationalism, shared identity, unifying institutions • Irredentism: claim territory based on ethnicity; Balkanization: fragmentation into smaller states (e.g., former Yugoslavia)	• Agriculture & geography: climate, soil, topography determine farming; subsistence (self-use) vs commercial (market-oriented) • Settlement: clustered (village), dispersed (isolated), linear (along road/river); survey: metes & bounds, township & range, long lot • Hearths: Fertile Crescent, Mesoamerica, SE Asia; 1st Ag Revolution (Neolithic) = plant/animal domestication • 2nd Ag Revolution: mechanization, crop rotation, selective breeding → food surplus → fueled Industrial Revolution • Green Revolution: high-yield seeds, fertilizers, irrigation in developing world; increased food but caused water depletion, soil degradation • Intensive: high labor/capital per acre — market gardening, plantation, mixed crop/livestock, paddy rice • Extensive: large area, low labor — shifting cultivation, pastoral nomadism, ranching, grain farming • Von Thünen Model: concentric land-use rings around market center; perishability + transport cost determine zones; modified by rivers, roads • Global system: commodity chains, agribusiness, fair trade; dependency on international markets; food deserts • Consequences: deforestation, desertification, soil erosion, salinization, water pollution, biodiversity loss • Challenges: GMOs, organic farming, aquaculture, urban farming, food sovereignty debates • Women in agriculture: primary subsistence producers in developing world; barriers to land ownership, credit, education	• Urbanization: city population growth; suburbanization = growth of suburbs; counterurbanization = movement to rural areas • Global cities: megacities (10M+); world cities (NYC, London, Tokyo) = economic command centers, finance/trade hubs • Rank-size rule: 2nd city = ½ of largest, 3rd = ⅓, etc.; primate city = disproportionately large (e.g., Paris) • Central place theory (Christaller): hexagonal market areas; higher-order services = wider spacing between centers • Gravity model: interaction between places proportional to population, inversely proportional to distance² • US city models: Burgess (concentric zone), Hoyt (sector), Harris-Ullman (multiple nuclei), galactic (peripheral/edge cities) • Non-US models: Latin American/Griffin-Ford (spine + sectors), SE Asian (port zone), African (3 CBDs), Islamic (mosque-centered) • Bid-rent theory: land value ↓ from CBD; commercial → residential → agricultural use outward • Zoning: separates land uses; edge cities = suburban business districts; exurbs = beyond suburbs • Infrastructure: transportation networks shape urban form; utilities, public services determine growth patterns • Urban sustainability: smart growth, new urbanism, mixed-use, greenbelts, transit-oriented development, LEED certification • Challenges: gentrification (displacement), sprawl, redlining, blockbusting; squatter settlements in developing world • Sustainability challenges: brownfields, pollution, congestion, urban heat island, food deserts, housing affordability



Unit 7 – Industrial & Economic Development

- **Industrial Revolution:** originated in Britain; steam power, mechanization, factory system → urbanization, labor migration
- **Economic sectors:** primary (extraction), secondary (manufacturing), tertiary (services), quaternary (info/research), quinary (high-level decisions)
- **Formal economy:** legal, taxed, regulated; **informal:** unregistered, untaxed — common in developing nations
- **Development measures:** GDP/GNI per capita, HDI (health + education + income), GII (gender inequality), literacy rate, life expectancy
- **Women & development:** GII tracks gender gaps; microloans empower women; education → economic participation
- **Rostow's modernization:** 5 stages (traditional → preconditions → takeoff → drive to maturity → mass consumption); critiqued as Western-centric
- **Wallerstein's world-systems:** core (wealthy, industrial), semi-periphery (transitioning), periphery (exploited, raw materials)
- **Dependency theory:** development of core exploits periphery; structural disadvantage perpetuated by unequal trade
- **Trade:** comparative advantage; WTO, IMF, World Bank; free trade vs protectionism; USMCA, EU, ASEAN
- **Economic changes:** deindustrialization, outsourcing, offshoring; SEZs, export processing zones; new international division of labor
- **Weber's Least Cost Theory:** factory location minimizes transport + labor + agglomeration costs; break-of-bulk points
- **Sustainable development:** balance economic growth with environmental protection; UN SDGs; appropriate technology



Key Models & Theories

- **Demographic Transition Model:** 5 stages of birth/death rate changes; explains population growth over time
- **Epidemiologic Transition Model:** disease cause-of-death shifts with each DTM stage
- **Malthusian Theory:** population outpaces food; critiqued — technology has increased food production
- **Ravenstein's Laws:** most migration is short-distance; step migration; migrants often young adults
- **Von Thünen Model:** agricultural land-use rings based on transport cost + perishability around a central market
- **Burgess Concentric Zone:** city grows outward in rings from CBD (1925)
- **Hoyt Sector Model:** city grows in wedge-shaped sectors along transportation routes (1939)
- **Harris-Ullman Multiple Nuclei:** city has several centers of activity, not just CBD (1945)
- **Galactic/Peripheral Model:** edge cities and suburban business nodes around central city
- **Griffin-Ford (Latin American):** CBD + spine + sectors; elite near center, disamenity at periphery
- **Central Place Theory** (Christaller): hexagonal market areas; threshold + range determine service locations
- **Gravity Model:** interaction ∝ (pop₁ × pop₂) ÷ distance²; predicts trade, migration, communication
- **Rank-Size Rule:** nth city = 1/n of largest city population; **primate city** = exception
- **Rostow's Modernization:** 5-stage linear development path; assumes all countries follow same trajectory
- **Wallerstein's World-Systems:** core/semi-periphery/periphery global power structure
- **Weber's Least Cost:** industrial location minimizes transport, labor, and agglomeration costs
- **Bid-Rent Theory:** land cost ↓ with distance from CBD; determines commercial/residential/agricultural zones
- **Dependency Theory:** periphery exploited by core; global trade perpetuates inequality



FRQ Tips & Exam Strategy

- **Exam format:** 60 MCQs (60 min, 50%) + 3 FRQs (75 min, 50%); FRQs are 7 points each
- **FRQ structure:** Q1 typically has a stimulus (map, graph, image); Q2–Q3 may or may not
- **Label every part** (A, B, C, etc.) clearly — graders look for specific answers by part
- **Task verbs matter:** Identify = name it; Define = what is it; Describe = how it works/looks; Explain = show cause-effect relationships; Compare = both similarities AND differences
- **Be specific:** use real-world examples — name countries, cities, policies, organizations, not vague "some place"
- **Keep it direct:** no introductions, no conclusions; restate the prompt and answer it immediately
- **Read stimuli carefully:** titles, captions, axes, legends, units; note patterns, outliers, and trends
- **One example = one example:** if asked for "one," give exactly one — extra won't earn credit and wastes time
- **Connect across units:** FRQs often combine concepts (e.g., migration + urbanization, agriculture + development)
- **Time management:** ~25 min per FRQ; don't spend too long on one — partial credit on all 3 beats perfection on 1